# Embedded Systems Tutorial #1 solutions

a) 0 = input 1 = output

| PB7 | PB6 | PB5 | PB4 | PB3 | PB2 | PB1 | PB0 |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |

= 0xEF hexadecimal

b) L6 and L7

c) R3 is a guard resistor in case PB4 is accidentally set as a high output which would cause a short circuit if no resistor were present.

d) There is no pull-up resistor to 5V, so the voltage is not going to be high when the switch is not pushed.

e) PORTB &= ~(1<<7);

f) PB0 = 1, PB2 = 0, PB3 = 0 (L1 & L3 on)
PB1 = 0 (L2 & L4 off)
PB6 = 1 PB7 = 1 (L6 & L7 off)
PB5 = 0 (L5 off)
PB4 X (input so will not affect the LEDs. If PORTx is activated when pin is set as an input, then the internal pull-up resistor will be enabled). We will use PB4 = 0 here

| PB7 | PB6 | PB5 | PB4 | PB3 | PB2 | PB1 | PB0 |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |

decimal value = 128 + 64 + 1 = 193

If PB4 = 1 instead
decimal value = 193 + 16 = 209

**Q2.** Data Input Output 2

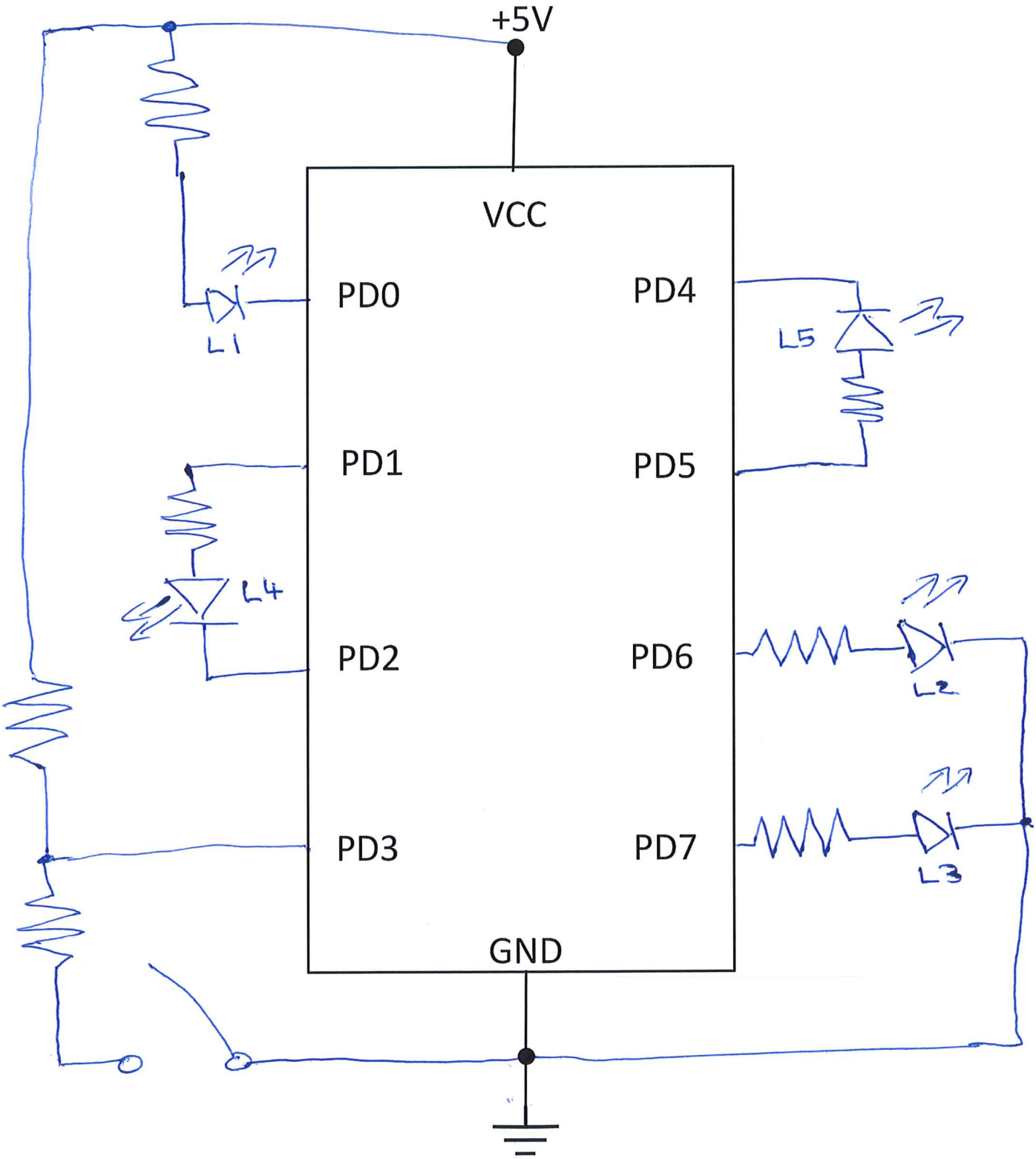



Figure 2: Schematic showing an ATmega32u2 microcontroller to be filled in for Question 2.

2.) a) See Figure 2.

b) 7 outputs, 1 input

input = 0, output = 1

input at bit # 3

binary: 1111 0111

decimal: 255 - 8 = 247

c) Active Low. Connected to LSB = PD0

See Figure 2

d) 0xC1 = 1100 0001

PD7 PD6

PD0 — turns off active low LED L1

L2 and L3 connected to PD6, PD7.

See Figure 2.

e) 35 = 32 + 2 + 1

binary 0010 0011

PD5

PD1 PD0 — active low LED L1

∴ PD1 and PD5 are connected to the anodes of L4 and L5. Adjacent pins are PD2 and PD4 respectively.